

Workspace-band diagnostics across medical SFT

Every probed checkpoint, gemma-3-{270m,1b,4b}-it

August 12, 2026

Full fine-tune on mix_default, lr $1e-5$, log-spaced checkpoints to step 4882. $t = 0$ is the untouched base.

Panels are medlens Figs. 27–28, rendered by band.band_diagnostic_plot. Within each section the y-axes and the CKA color scale are **pinned across all steps**, so what moves on the page is the metric, not the axis.

How to read the panel

CKA among J-lens vectors — linear CKA between layers over a subsample of meaningful vocabulary tokens. Block structure exposes the sensory / workspace / motor split.

Next-token prediction accuracy — fraction of positions where any top-10 lens token matches the model's own top-1. Low in the band (read-outs carry forward/abstract content), igniting in the motor layers.

Excess kurtosis — of the lens-logit distribution over the vocabulary. Near zero early, rising once read-outs carry real content.

Top-1 autocorrelation — $P(\text{top-1 identical at adjacent positions})$ as $\Delta \log p$ against a position-shuffled null. High when the J-space carries content that persists across positions.

J-space dimensionality — fraction of residual-stream dimensions needed for 90% of the variance across the J-lens vectors.

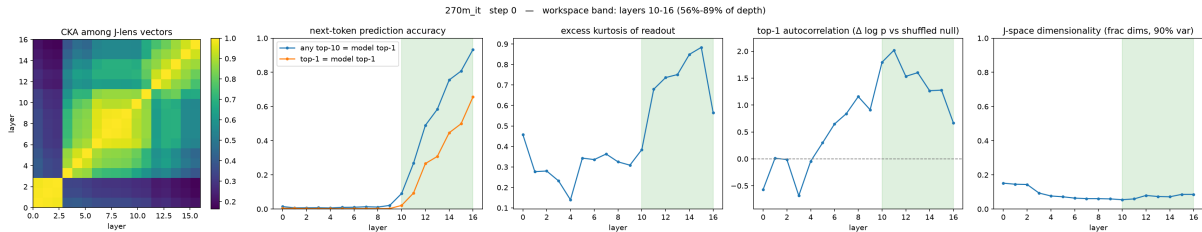
Green shading is `locate_band`'s *suggested* contiguous band — a suggestion generator, **not human-confirmed**. Onset is a threshold on kurtosis and autocorrelation excess, so it can flip on a shallow curve; read the curves, not the shading.

270m-it

18 layers · 13 checkpoints + $t=0$

band **never moved**: 10–16 at every step

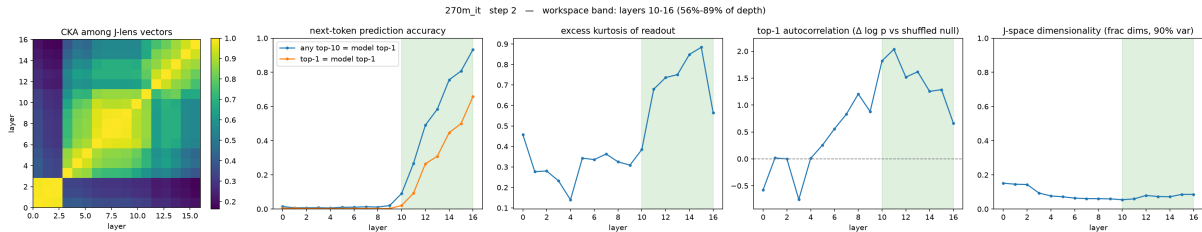
270m-it — $t=0$ (untuned base)



band: layers **10–16** (56%–89% of depth)

trajectory origin

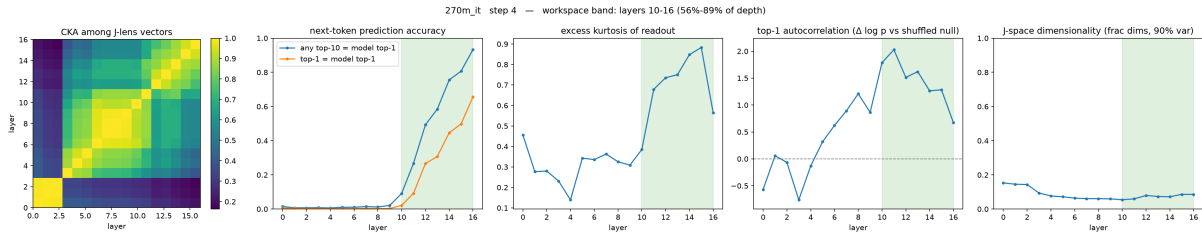
270m-it — step 2



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

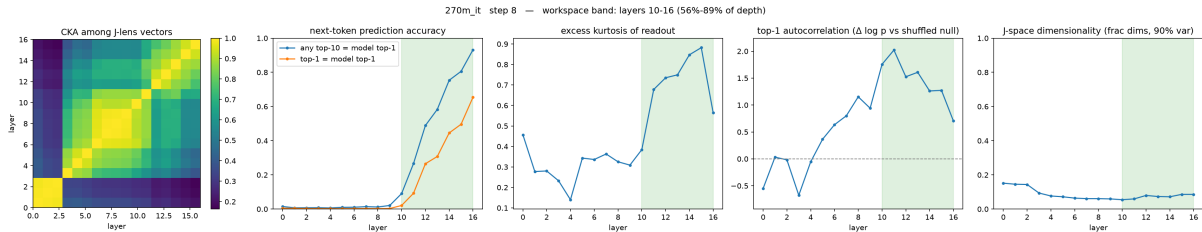
270m-it — step 4



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

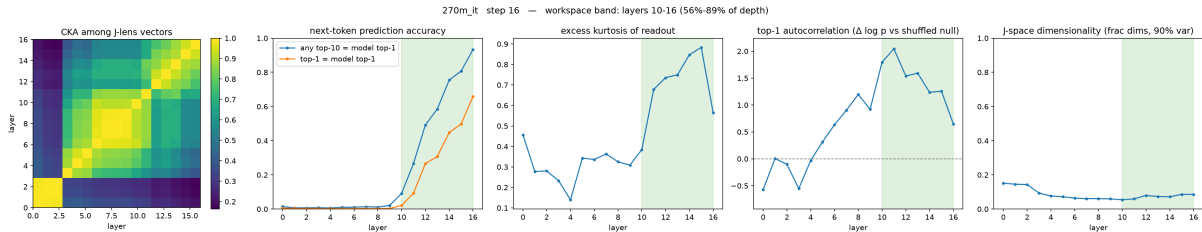
270m-it — step 8



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

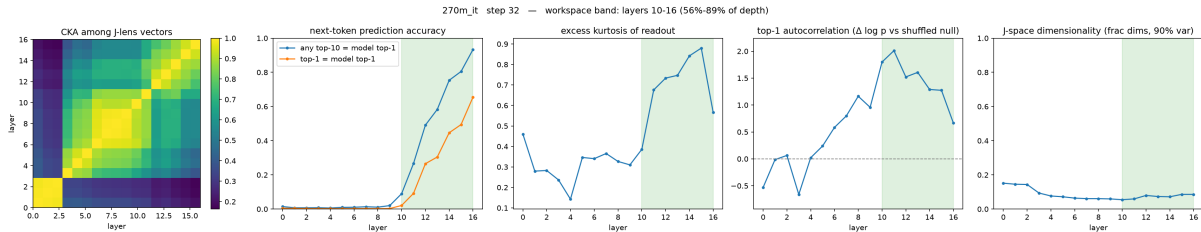
270m-it — step 16



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

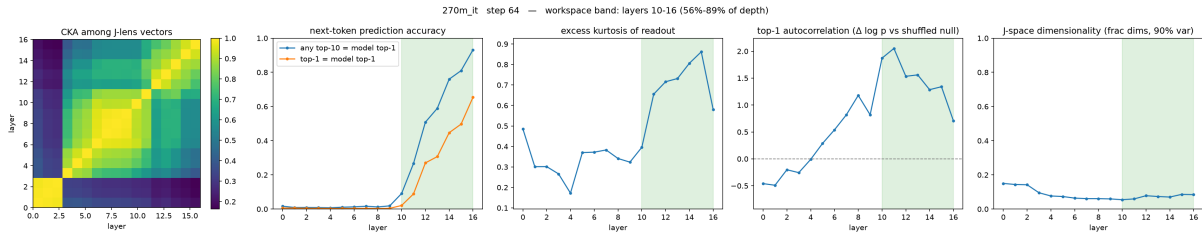
270m-it — step 32



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

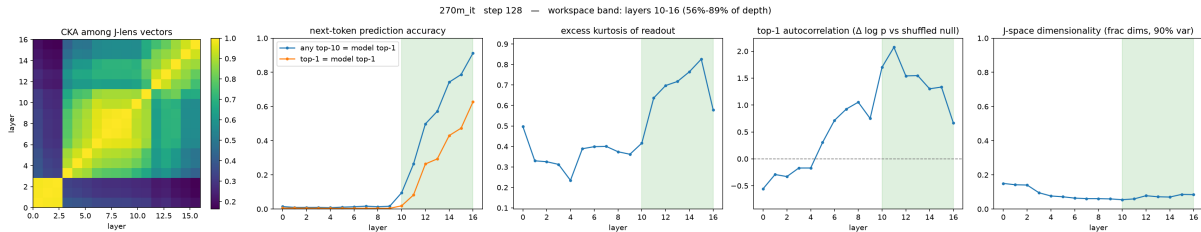
270m-it — step 64



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

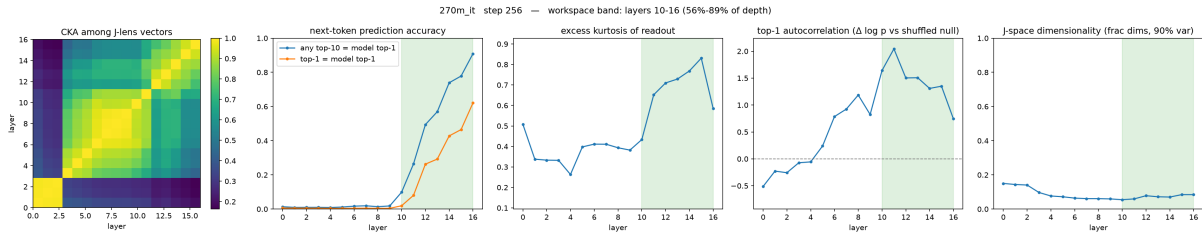
270m-it — step 128



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

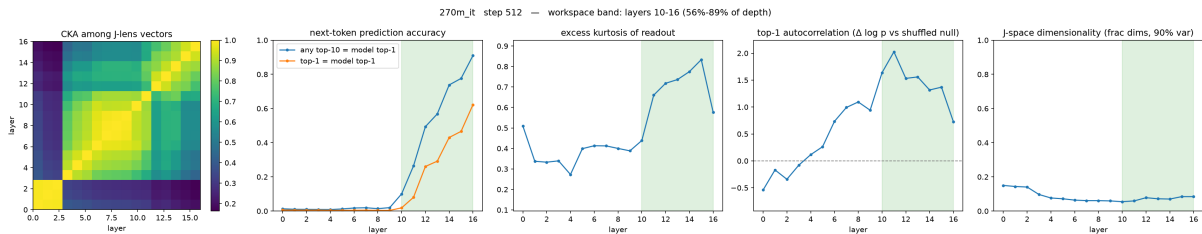
270m-it — step 256



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

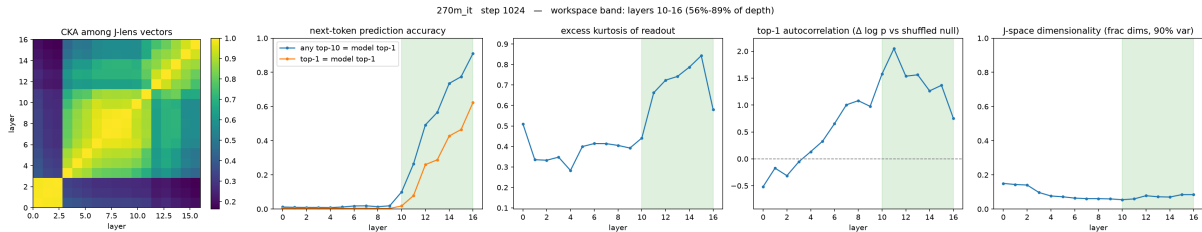
270m-it — step 512



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

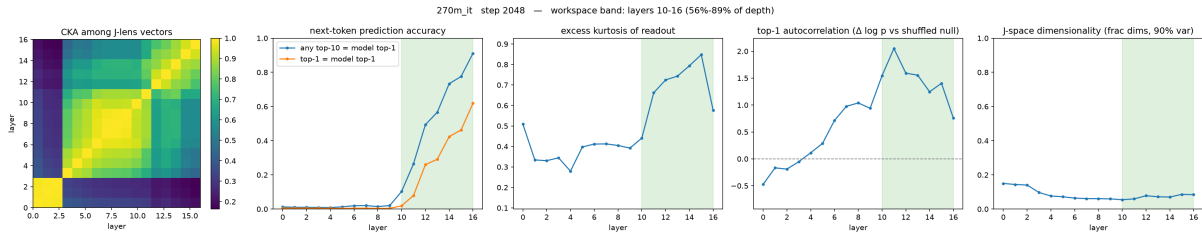
270m-it — step 1024



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

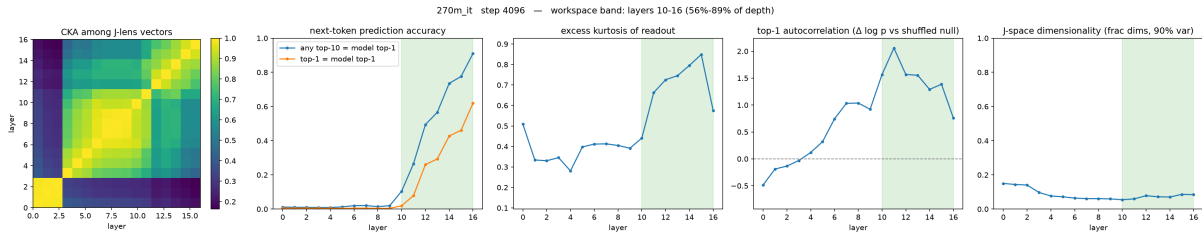
270m-it — step 2048



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

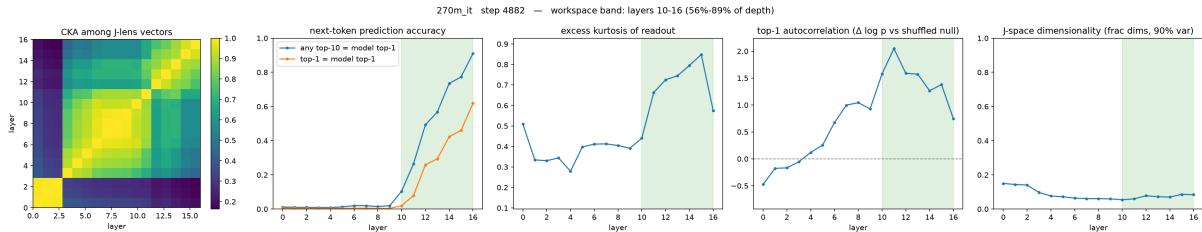
270m-it — step 4096



band: layers 10–16 (56%–89% of depth)

unchanged from previous step

270m-it — step 4882 (final)



band: layers 10–16 (56%–89% of depth)

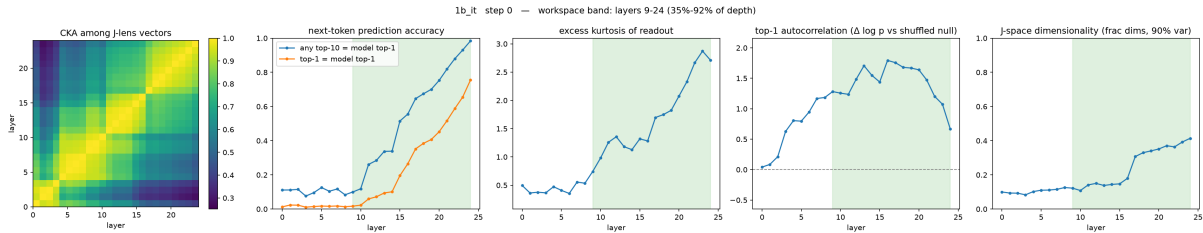
unchanged from previous step

1b-it

26 layers · 14 checkpoints + $t=0$

band moved 9-24 → 14-24

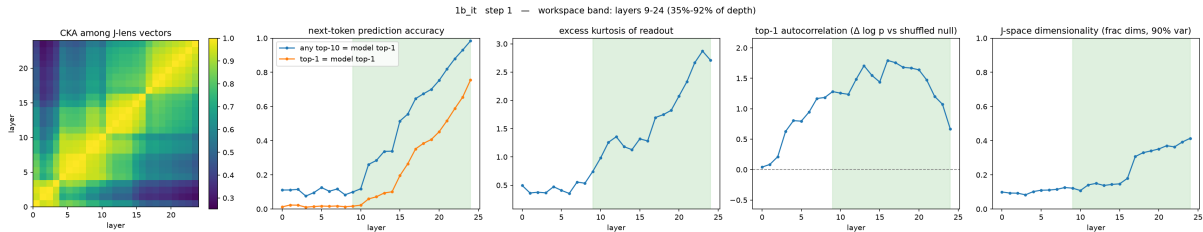
1b-it — $t=0$ (untuned base)



band: layers **9–24** (35%–92% of depth)

trajectory origin

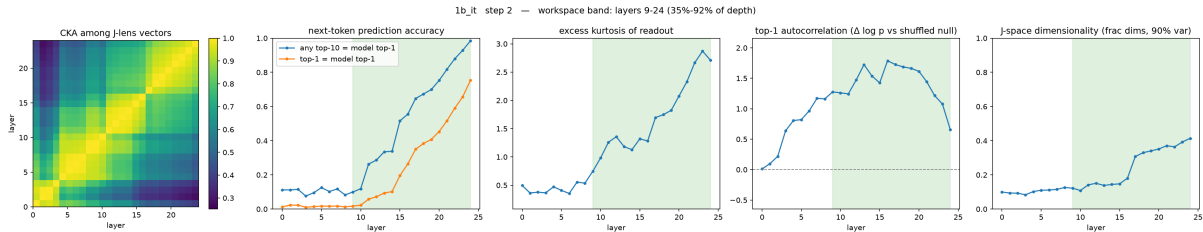
1b-it — step 1



band: layers **9–24** (35%–92% of depth)

unchanged from previous step

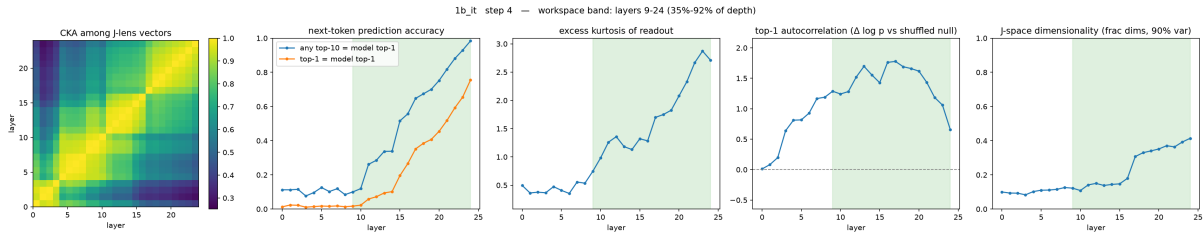
1b-it — step 2



band: layers **9–24** (35%–92% of depth)

unchanged from previous step

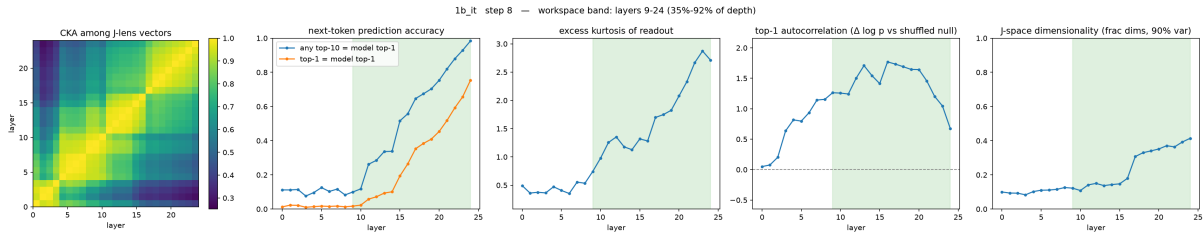
1b-it — step 4



band: layers **9–24** (35%–92% of depth)

unchanged from previous step

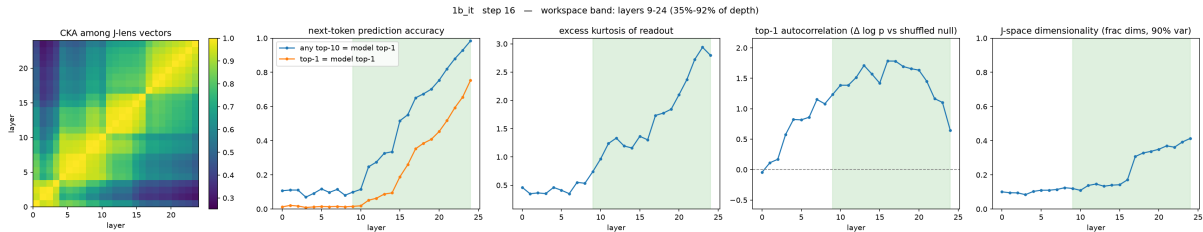
1b-it — step 8



band: layers **9–24** (35%–92% of depth)

unchanged from previous step

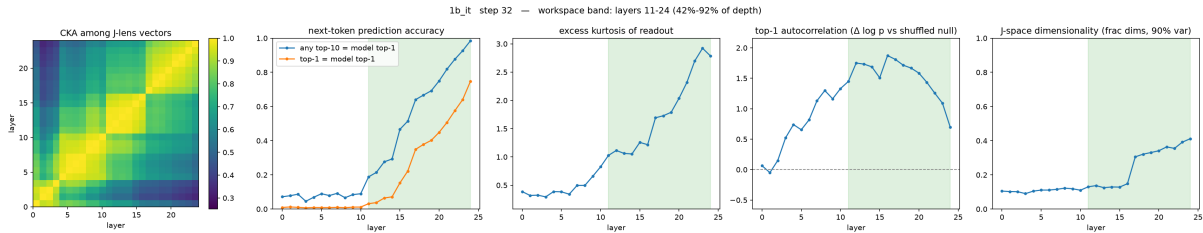
1b-it — step 16



band: layers **9–24** (35%–92% of depth)

unchanged from previous step

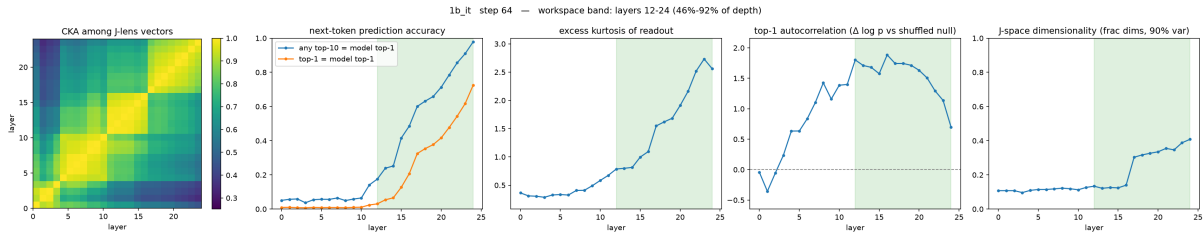
1b-it — step 32



band: layers **11–24** (42%–92% of depth)

onset 9 → **11** (up)

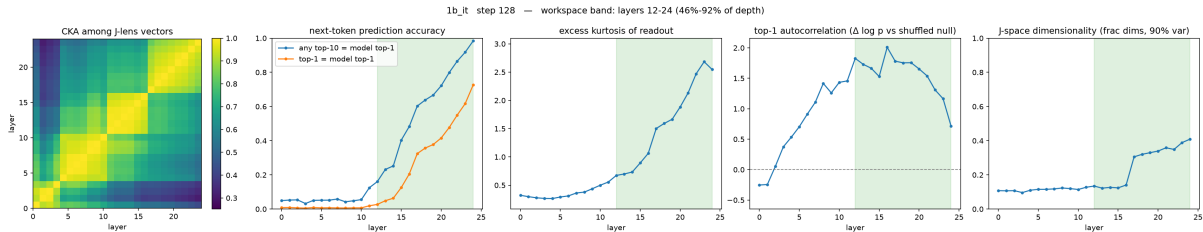
1b-it — step 64



band: layers 12–24 (46%–92% of depth)

onset 11 → 12 (up)

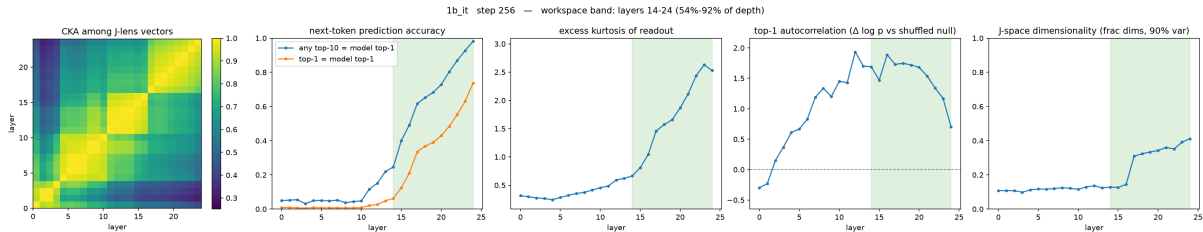
1b-it — step 128



band: layers 12–24 (46%–92% of depth)

unchanged from previous step

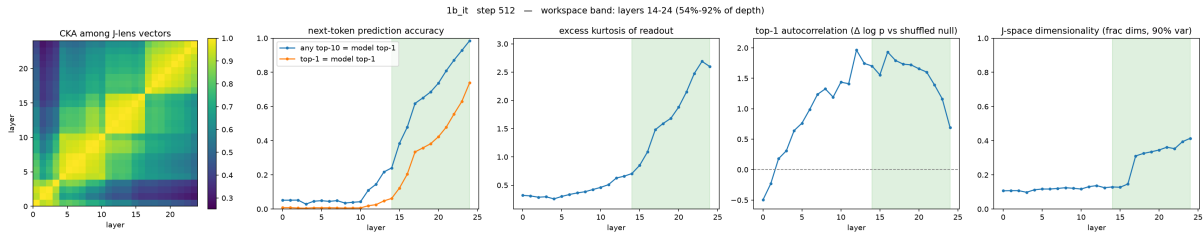
1b-it — step 256



band: layers **14–24** (54%–92% of depth)

onset 12 $\rightarrow$ **14** (up)

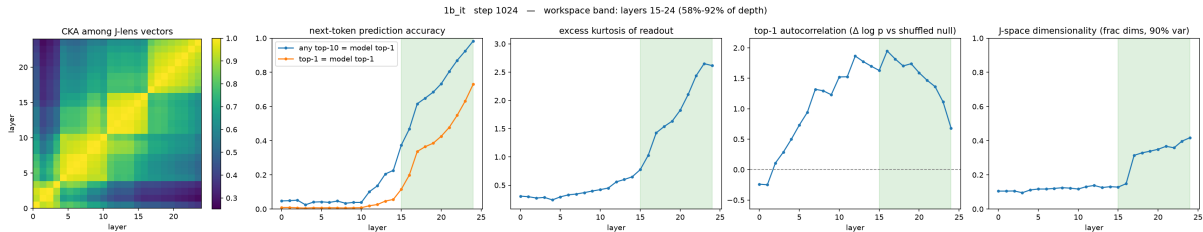
1b-it — step 512



band: layers 14–24 (54%–92% of depth)

unchanged from previous step

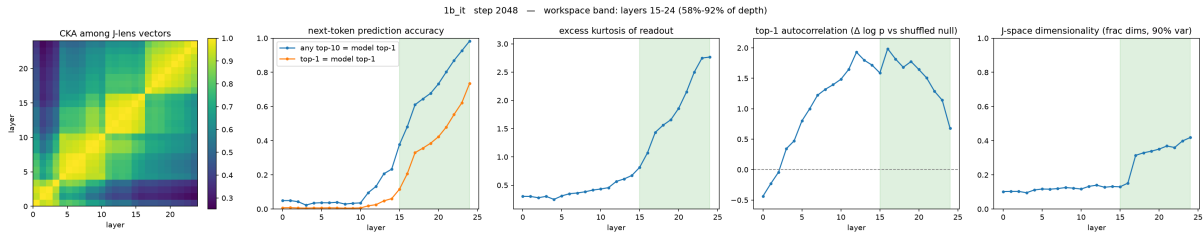
1b-it — step 1024



band: layers 15–24 (58%–92% of depth)

onset 14 → 15 (up)

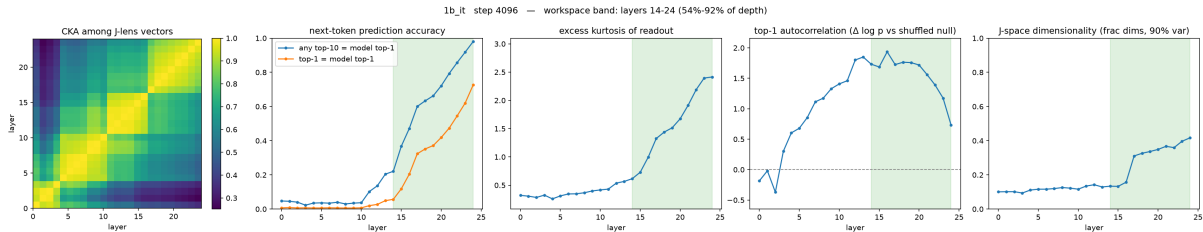
1b-it — step 2048



band: layers 15–24 (58%–92% of depth)

unchanged from previous step

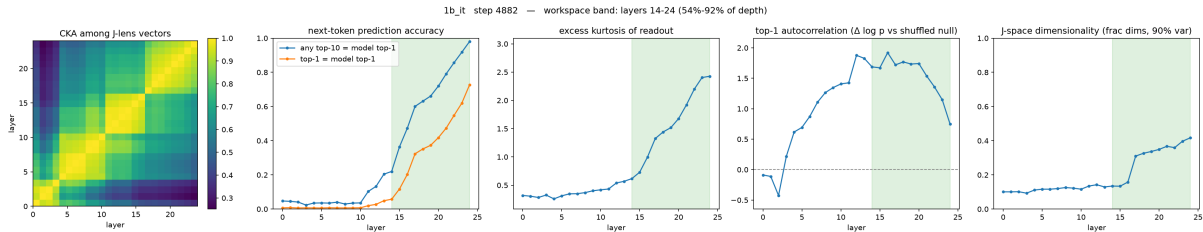
1b-it — step 4096



band: layers **14–24** (54%–92% of depth)

onset 15 $\rightarrow$ **14** (down)

1b-it — step 4882 (final)



band: layers **14–24** (54%–92% of depth)

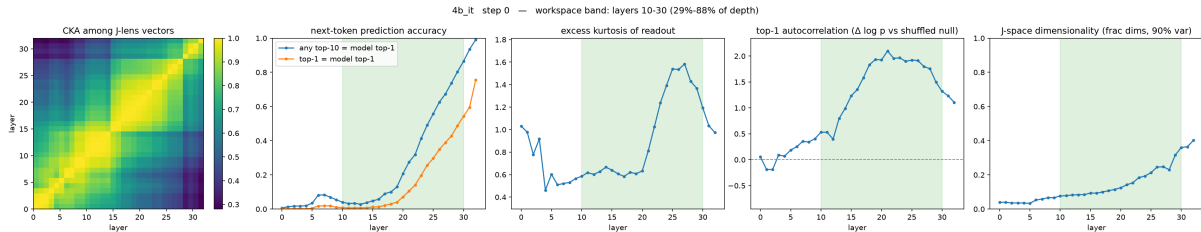
unchanged from previous step

4b-it

34 layers · 13 checkpoints + $t=0$

band moved 10-30 → 9-30

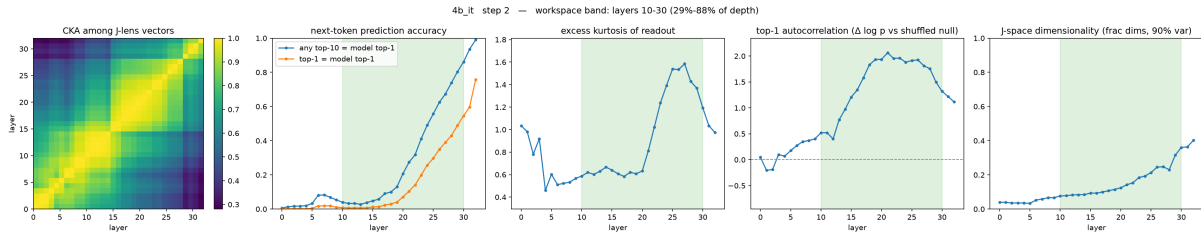
4b-it — $t=0$ (untuned base)



band: layers 10–30 (29%–88% of depth)

trajectory origin

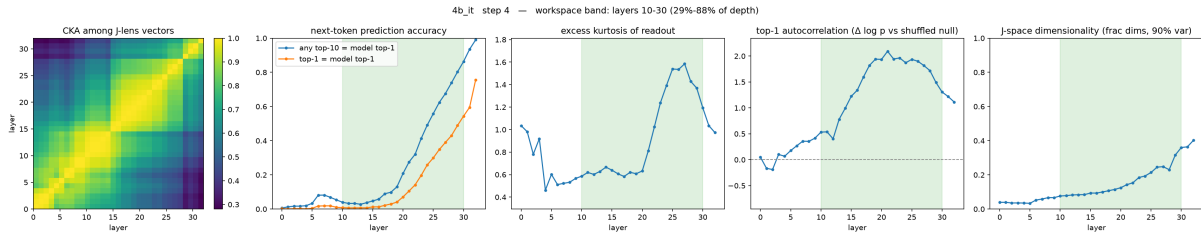
4b-it — step 2



band: layers 10–30 (29%–88% of depth)

unchanged from previous step

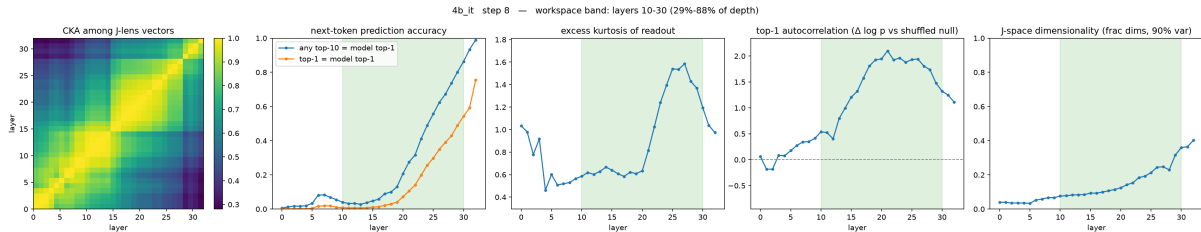
4b-it — step 4



band: layers 10–30 (29%–88% of depth)

unchanged from previous step

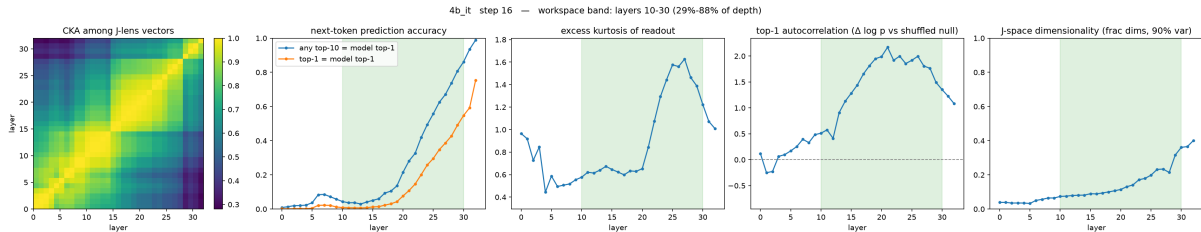
4b-it — step 8



band: layers 10–30 (29%–88% of depth)

unchanged from previous step

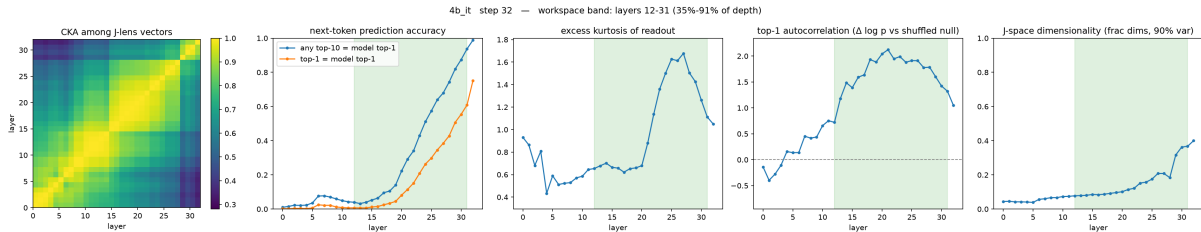
4b-it — step 16



band: layers 10–30 (29%–88% of depth)

unchanged from previous step

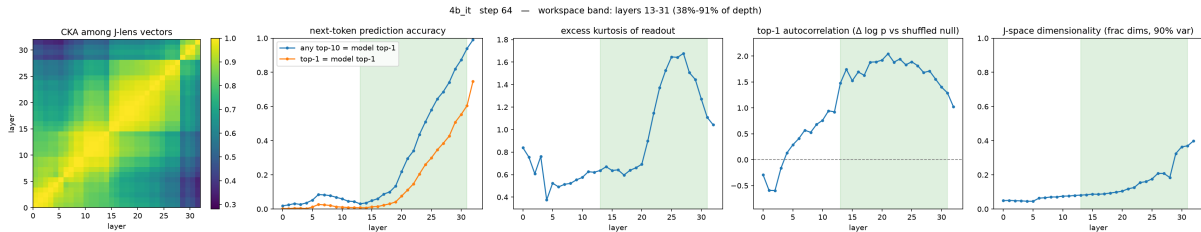
4b-it — step 32



band: layers **12–31** (35%–91% of depth)

onset 10 $\rightarrow$ **12** (up), offset 30 $\rightarrow$ **31** (up)

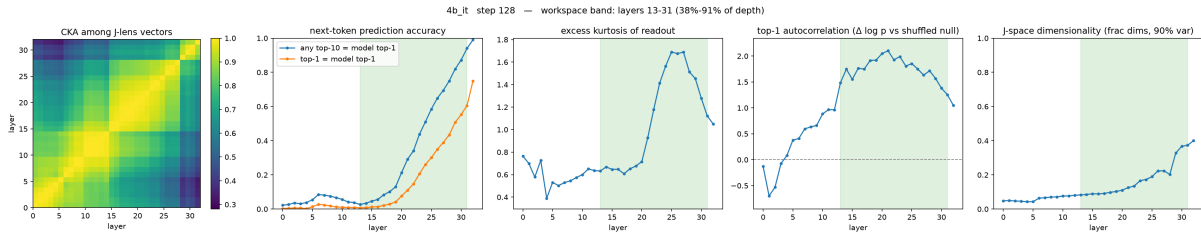
4b-it — step 64



band: layers **13–31** (38%–91% of depth)

onset 12 $\rightarrow$ **13** (up)

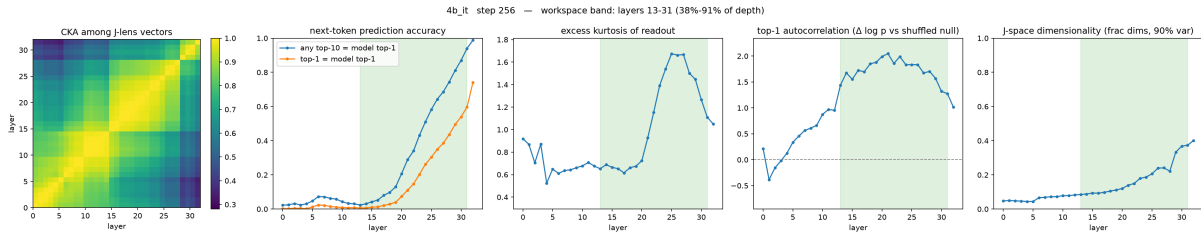
4b-it — step 128



band: layers **13–31** (38%–91% of depth)

unchanged from previous step

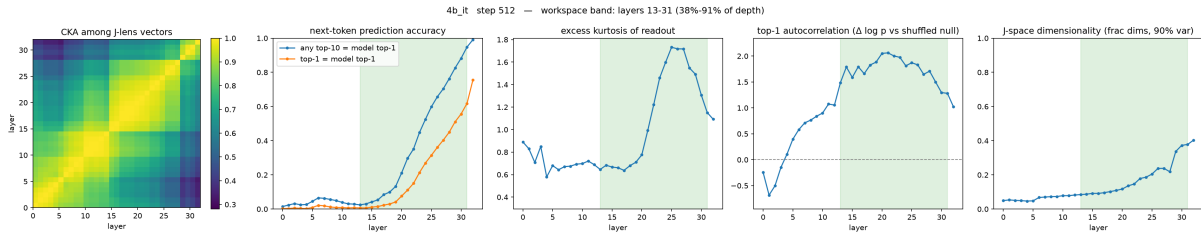
4b-it — step 256



band: layers **13–31** (38%–91% of depth)

unchanged from previous step

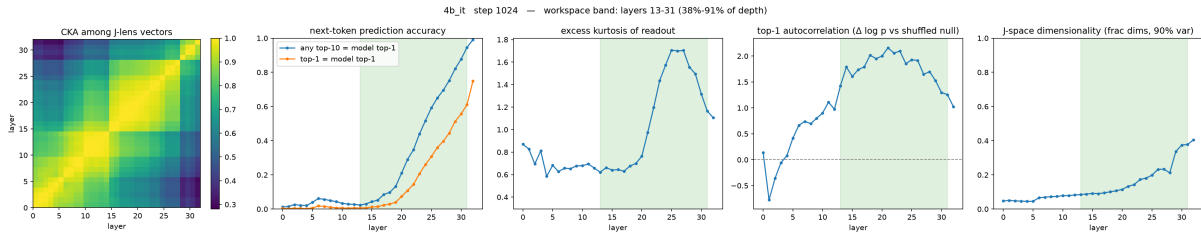
4b-it — step 512



band: layers **13–31** (38%–91% of depth)

unchanged from previous step

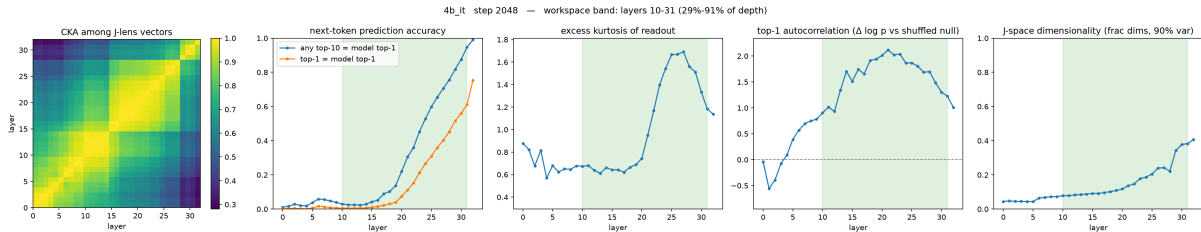
4b-it — step 1024



band: layers **13–31** (38%–91% of depth)

unchanged from previous step

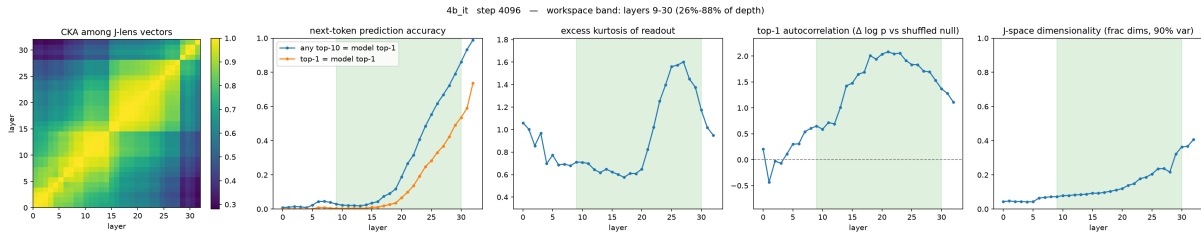
4b-it — step 2048



band: layers **10–31** (29%–91% of depth)

onset 13 → **10** (down)

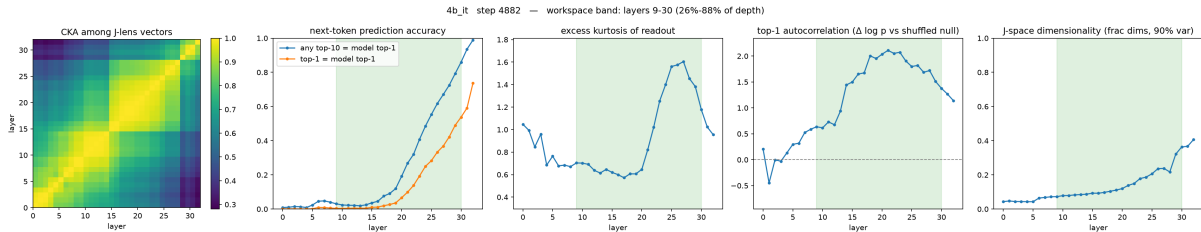
4b-it — step 4096



band: layers **9–30** (26%–88% of depth)

onset 10 $\rightarrow$ **9** (down), offset 31 $\rightarrow$ **30** (down)

4b-it — step 4882 (final)



band: layers **9–30** (26%–88% of depth)

unchanged from previous step